

# AYUSH NATHANI

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## Education

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|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| <b>Indian Institute of Technology (IIT) Patna and Indian Institute of Information Technology (IIIT) Ranchi</b><br><i>Master of Computer Application (MCA) (CGPA: 10.0)</i> | <b>Jul 2025 – Present</b><br><i>Bihar, India</i>      |
| <b>Amity University Jharkhand</b><br><i>Bachelor of Computer Applications (BCA) (CGPA: 8.72)</i>                                                                           | <b>Jul 2021 – Jun 2024</b><br><i>Jharkhand, India</i> |

## Technical Skills

- Languages:** Python, JavaScript, TypeScript
- Frameworks:** FastAPI, React, Next.js, Express.js
- Platforms:** MongoDB Atlas, Vercel, Google Cloud, Railway, Microsoft Azure, Render, Netlify, Firebase
- Tools:** Git, GitHub, Docker

## Projects

### TurfGrid AI | Autonomous Smart City Command Center Jun 2026

- Architected a **5-agent autonomous smart city platform** (Orchestrator + Fan, Business, Crowd, Operations sub-agents) **managing fan travel, venue operations, and crowd control** across 23 FIFA World Cup 2026 venues in 3 countries.
- Implemented **MongoDB Atlas \$vectorSearch** with 768-dim **Gemini text-embedding-2 vectors** for semantic venue discovery via cosine similarity, enabling natural-language queries like 'stadiums near water with large capacities'.
  - Engineered high-availability LLM failover: **Gemini 2.5 Flash** as primary orchestrator with seamless rerouting to **Groq Llama-3.3-70b** on resource exhaustion, preserving full tool-calling parity and multi-turn context.
  - Built true **agent-to-agent orchestration** where Fan Agent itinerary saves auto-trigger Business Readiness Agent workflows, with all inter-agent actions logged to MongoDB and surfaced as real-time Workflow Toasts in the UI.

[Demo Link](#) | [Python, FastAPI, Next.js, MongoDB Atlas Vector Search, Gemini 2.5 Flash, Groq Llama-3.3-70b, FastMCP]

### ReelForge AI | Autonomous AI Video Generation Platform Mar 2026

- Architected a **6-agent autonomous content pipeline** (Script Writer -> Scene Director -> Video Producer -> Audio Producer -> Viral Optimizer -> Renderer) converting text prompts into production-ready short-form videos.
- Integrated **multimodal AI services** (RunwayML, ElevenLabs) for Text-to-Image, Image-to-Video, Text-to-Speech, and AI sound effect workflows in a unified pipeline with **real-time SSE progress monitoring**.
  - Engineered **serverless FFmpeg rendering pipeline** for scene concatenation, narration sync, audio normalization, and **automated MP4 export with an AI viral analysis** engine scoring hook quality and emotional resonance.

[Demo Link](#) | [Next.js, React, TypeScript, RunwayML, ElevenLabs, FFmpeg, Server-Sent Events (SSE)]

### Email Triage AI | Autonomous Multi-Agent Email Management System Nov 2025

- Built a **production-grade multi-agent email automation platform** autonomously classifying, prioritizing, routing, and drafting responses for live **Gmail inboxes using Groq low-latency LLM inference**.
- Designed a **3-stage agentic pipeline** (Classifier -> Responder -> Router) with confidence-based scoring, explainable chain-of-thought reasoning, and tone-adaptive response generation.
  - Implemented **automated PII redaction layer** before LLM inference, **containerized on Hugging Face Spaces via Docker** with real-time dashboard tracking agent decisions, routing paths, and confidence metrics.

[Demo Link](#) | [Python, FastAPI, Groq (Llama 3.1-8b-instant), Gmail API, OAuth2, Docker, Pydantic]

## Certifications

- Microsoft Azure AI Essentials: Workloads and Machine Learning on Azure:** Introduction to Azure AI workloads and basic machine learning concepts. [Certificate](#)
- Gemini for Google Workspace:** Practical application of Gemini AI tools for Google Workspace. [Certificate](#)